

Hugo Garrido-Lestache

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SUMMARY

- Passionate about Researching in AI and Reinforcement Learning, Aspiring Graduate Student and Published Author attending ICMLA!
- Support me on YouTube at [@HugoDoesAi](#) and come checkout my [WEBSITE](#).

EDUCATION

M.S. Machine Learning	Milwaukee School of Engineering	August 2026
B.S. Computer Science (GPA: 3.95)	Milwaukee School of Engineering	Winter 2025
Game Design & Game Theory Course	University of Lucca	June 2023

RESEARCH EXPERIENCE

Undergrad Research Assistant | ICMLA 2025 January 2025 – Present

Advisor: [Jeremy Kedziora, Ph.D.](#)

[Paper](#) | [Blog](#) | [Talk](#) | [Code](#)

- Developing TAAC a Reinforcement Learning Algorithm to promote Collaboration using Attention in Multi-Agent Environments.
- Got Promising Performance on a soccer environment and tower building environment.
- Presented at a computing symposium and published a paper as first author.
- Came 1st in the MSOE NVIDIA ROSIE Super-Computer Competition winning over \$9,000 in prizes.

RESEARCH PROJECTS

Research on Silent Video Sound Generation | MICS 2024

Collaborators: Adam Haile & Helina Mulugeta

[Paper](#)

- Researched sound synthesis for silent videos using VQ-VAE.

Research on Developing Education AI Tool

Collaborators: Ethan Jeffers & Ethan Wrasman

[Paper](#)

- Developed AI tools to enhance learning in early programming Education through LLMs.
- Statistically tested these tools to ensure it effectiveness and prevent negative consequences.

WORK EXPERIENCE

Data Science Intern | Direct Supply

November 2023 – Present

- Ideated and Tested AI tools to solve problems.
- **AI Product Web Scraper:** Created an AI tool to search, parse and combine unstructured data on the internet to augment the company's quality and quantity of data.
- **AI Product Linking:** Improved AI systems to match products together based on semantic similarity.
- **AI Product Tagging:** Created a System to add tags/attributes to products automatically based on product data. This data can be used to group and filter products and feed future AI Models.

PROJECTS

Wordle Solver AI

[Blog](#) | [Code](#)

- Created an AI solver using Information Theory which achieves an average of 3.8 guesses per game.
- Made optimizations to allow for faster computer on mobile device.
- Created a test suite to test performance with various changes to allow fast iteration.

Snake Game Reinforcement Learning AI

[Blog](#) | [Code](#)

- Created a Q-learning AI agent to learn to play the game achieving a high score of 140.
- Tested out ideas to improve results to such as simplifying state + action space and tuning rewards and hyperparameters

Large Language Model - HugoGPT

[Blog](#) | [Code](#)

- Implemented an LLM from scratch and trained it on a supercomputer.
- Got results which looked like English but failed to have coherent sentences due to lack of data, hardware and time.

Software Development Project, Helping Kids LTD

- Developed a program for storing and searching therapist client data
- Implemented login systems with hashing and MySQL Database

MENTORING & ACTIVITIES

AI Club Mentor | MSOE AI club

Oct 2024 – Present

- Mentoring AI members on AI concepts and algorithms.
- 2025 Member's ended up creating their own CNN on their selected datasets.

Programming Tutor | Milwaukee School of Engineering

September 2023 – December 2023

- Working weekly with around 20 different students to explain topics and concepts in Programming.

Adult Scout Leader | UK Scouts | September 2020 - June 2022

Tau Beta Pi Member | Tau Beta Pi | April 2025 - Present

AI Club Member | AI Club | September 2022 - Present

Scouts | UK Scouts | 2014 - 2021

Climber | Climbing Club | September 2019 - Present

AWARDS & STUFF

1st in NVIDIA ROSIE Supercomputer Challenge | NVIDIA | [LINK](#)

6x Dean's List with High Honors Award | Milwaukee School of Engineering | [LINK](#)

Tau Beta Pi Member | Tau Beta Pi | April 2025 - Present

Mr. MSOE 2023 | Society of Women Engineers

5th in MICS Programming Competition | Midwest Instruction and Computing Symposium

Fundamentals of Deep Learning | NVIDIA | [LINK](#)

Silver Duke of Edinburg Award | Duke of Edinburg

A* - Maths & A* - Computer Science & A – Physics | English A-Levels

INTERESTS

Guitar | Climbing | Puzzles | Yo-yo | Origami | Unicycle | Cooking | Gaming | Youtube